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Automotive Cyber Security Academy 2023

Chapter 4 – Automotive Hardware Security

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Trusted Platform Module

Trusted Computing

- Trusted Computing Group (TCG) standardizes Trusted Computing (TC) concepts
  - Industrial and research consortium founded in 2003
  - Goal: development of open, platform- and manufacturer-independent standards for TC building blocks
  - Trusted Platform Module (TPM) is just one technology
- Goals of TC
  - Proof that security mechanisms are correctly implemented and not manipulated

https://trustedcomputinggroup.org/

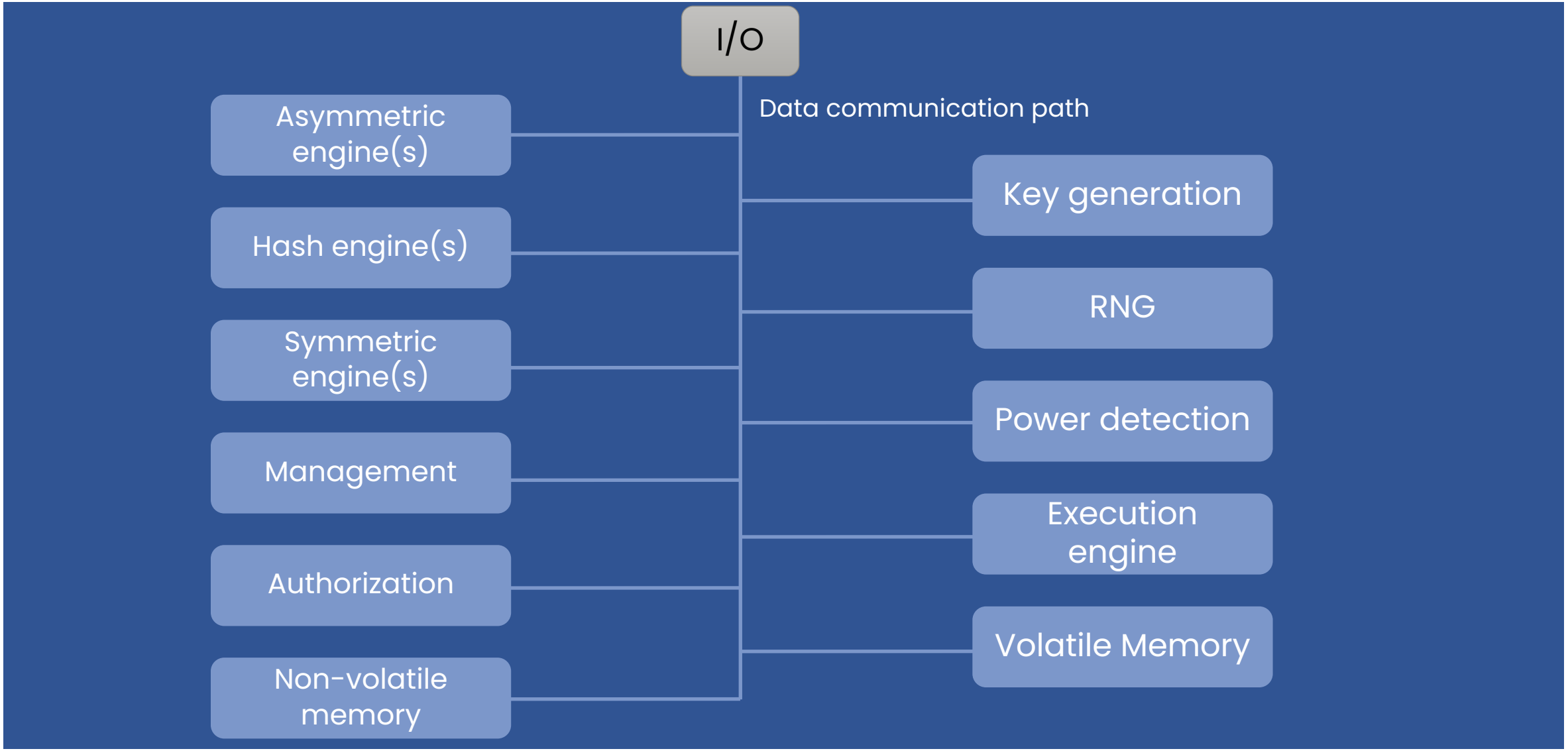
TPM Overview

- TPM is a type of HSM
- Two Specifications: TPM 1.2 and TPM 2.0
  - Specifications are not compatible
  - TPM 2.0 is a total rewrite and suitable also for embedded systems
- TPM library specification allows library profiles to meet different requirements
  - E.g., resource constraints or required functionalities
- TPM library profiles are specifications with
  - (Usually) reduced functionalities
  - Dedicated requirements for implementation

Key Features TPM 2.0

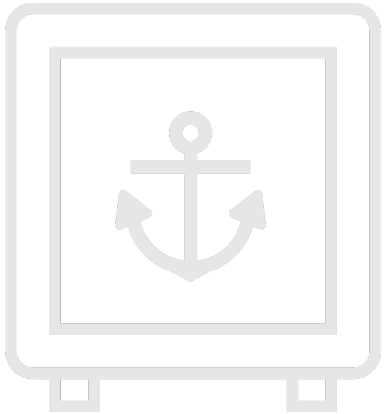
- Device identification
- Secure generation, storage, and usage of keys
- Root of trust for storage, measurement, and **reporting***
- **Cryptographic agility**
- Enhanced authorization
- Flexibility (TPM library profiles, dedicated hardware chip vs. firmware TPM …)
- Monotonic counters
- *Remote attestation

TPM 2.0 Architecture



TPM 2.0 Roots of Trust

- TCG defines three types of Roots of Trust
  - Root of Trust for Storage (RTS)
    - Storage of measurements of the system integrity
    - Protects measurements against unauthorized access
  - Root of Trust for Measurement (RTM)
    - Sends integrity information (i.e., measurements) to RTS
    - Controlled by Core Root of Trust for Measurement (CRTM)
    - CRTM is the initial boot code executed first on startup and initiates the measurement chain
  - Root of Trust for Reporting (RTR)
    - Reporting of measurements to external entities
    - Ensures integrity and authenticity of the reported data
    - TPM basically signs this data



Roots of Trust **must be trustworthy** since manipulations cannot be detected

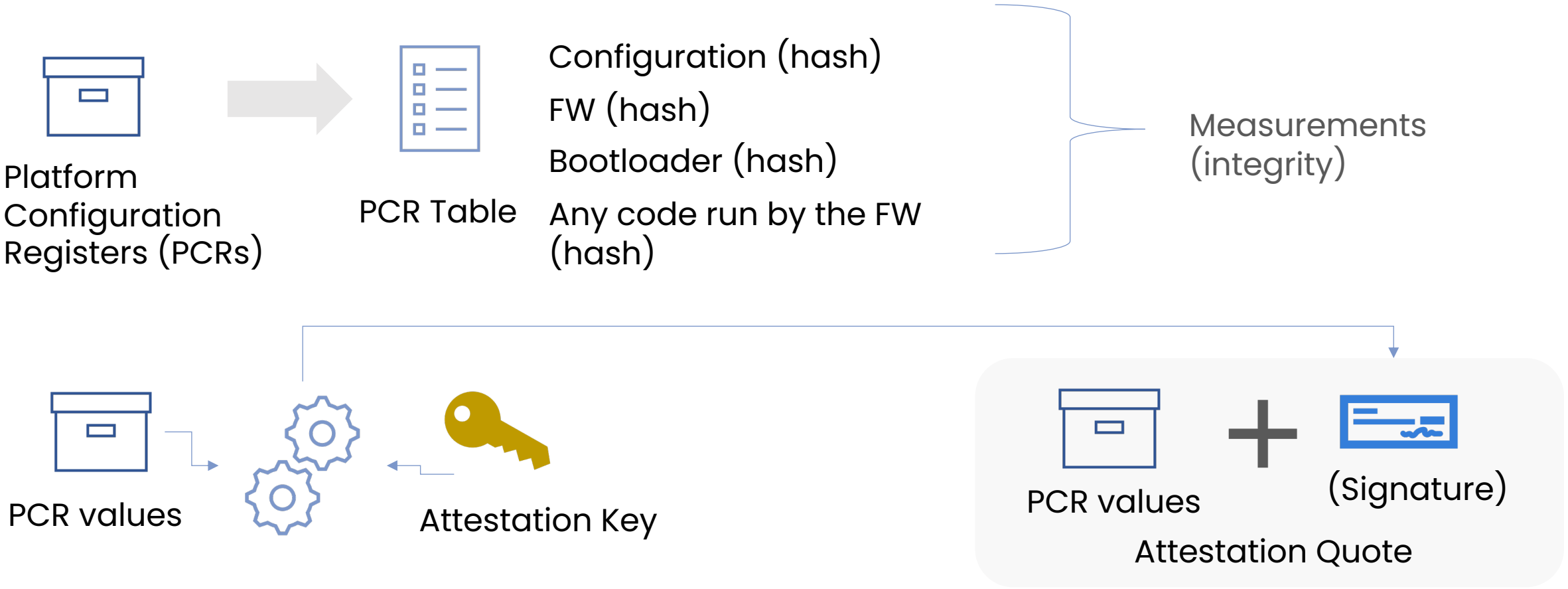
TPM 2.0 Crypto Keys

- Keys can be generated from a seed, generated using the TPM's RNG, or imported
- Key use can be controlled, e.g., restricted, sign, decrypt
- Support of different key types, e.g.
  - Storage (restricted decryption keys): encryption of data and other keys of the hierarchy
  - Certifying (Attestation keys): signature of TPM quotes and TPM credentials
  - Signing: signing of arbitrary data, e.g., certificates
  - Sealing
    - Encryption / decryption of small data (e.g., symmetric encryption keys)
    - Can be bound to platform configuration

TPM 2.0 Identity

- TPM Identity
  - Each TPM has a unique identity: Endorsement Key (EK) pair
    - Private key never leaves the TPM
    - Public key leaves the TPM
  - EK usually generated by the manufacturer and certified
    - Endorsement Credential (EC): x.509 v3 certificate
    - EC mostly stored in NVRAM of the TPM
  - EK and EC can be used to verify the correct identity of the TPM
- Platform Identity
  - TPM can be used as RTR
  - Ensure that integrity measurements belong indeed to the host platform
  - Platform Attribute Credential assures physical binding between TPM and host platform
    - Attests that a specific platform contains a unique TPM and Trusted Building Block (TBB), i.e., the Core Root of Trust for Measurement (CRTM) and the TPM initialization functions

TPM 2.0 Remote Attestation



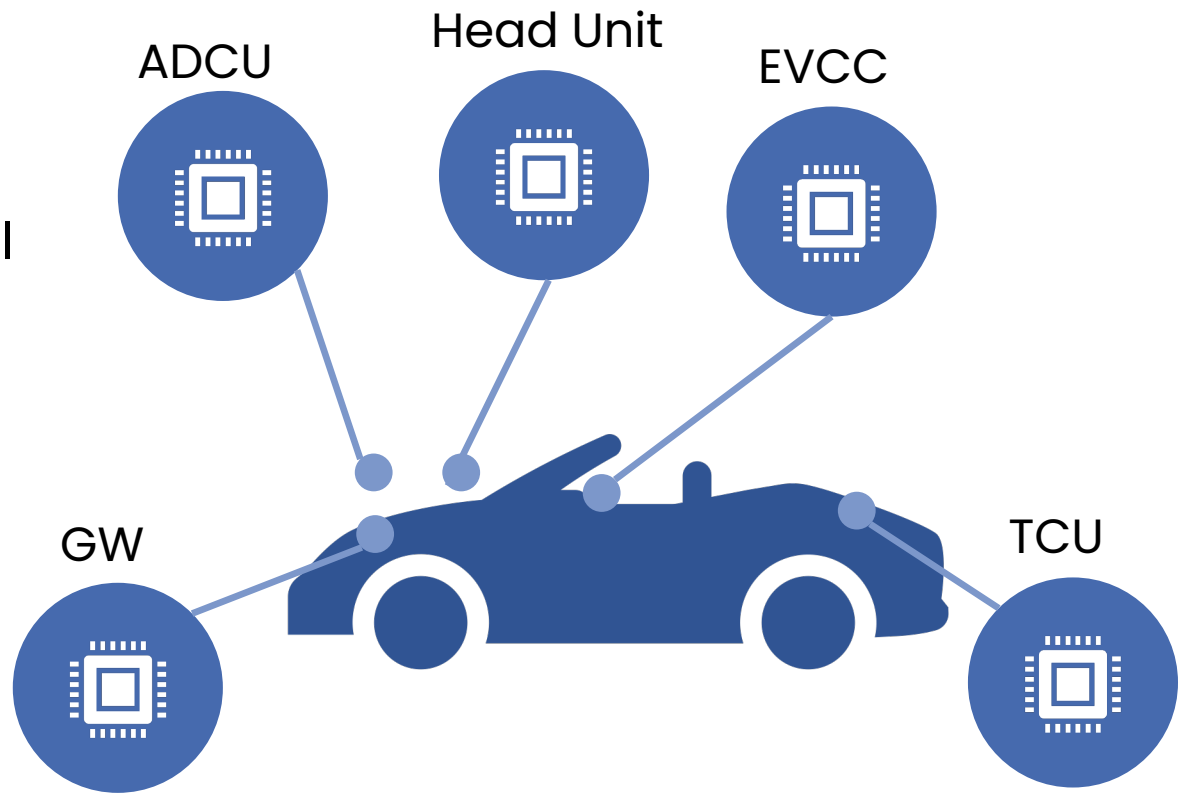
TPM 2.0 Library Profiles

TPM 2.0 Automotive Thin Profile

- PC Client
  - TPM for PC systems
  - Contains virtually the entire TPM 2.0 specification
- **Automotive Thin**
  - Very reduced TPM for automotive with low costs
  - Reduced set of supported cryptographic algorithms and TPM commands
- Mobile Common
  - For mobile devices, e.g., smartphones, tablets
  - Firmware TPM with dedicated implementation requirements

TPM 2.0 Automotive Integration

- Vehicle electrical systems sometimes consist of 100 Electronic Control Units (ECUs)
- Not possible to equip each ECU with a TPM
- A proposal: 5 TPMs in a vehicle
  - Basic setting: only one TPM, e.g., in the central gateway acts as a central key server
  - Extended settings integrate TPMs, e.g., in
    - Electric Vehicle Communication Controller (EVCC)
    - Head Unit
    - Telematics Control Unit (TCU)
    - Automated Driving Control Unit (ADCU)
- Other ECUs should use other solutions



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Thank You

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